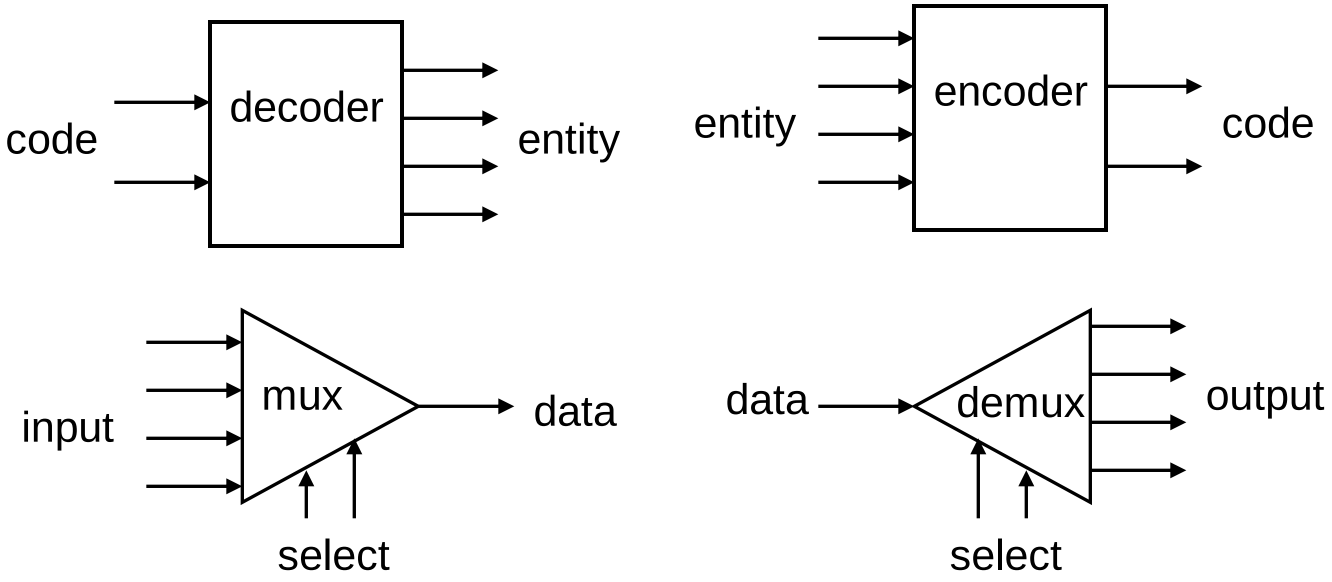
MSI Circuits

Useful MSI circuits

- Four common and useful MSI circuits are:
  - Decoder
  - Demultiplexer
  - Encoder
  - Multiplexer
- Block-level outlines of MSI circuits:

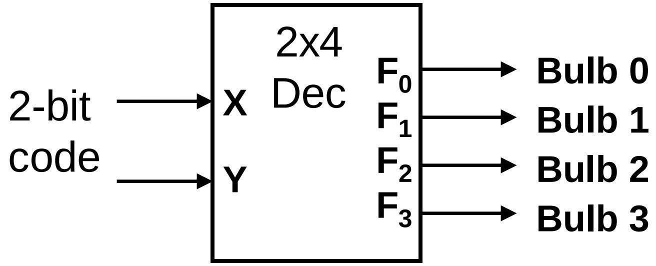


Decoders

- Convert binary information from n input lines to (max. of) 2^n output lines.
- Known as n-to-m-line decoder, or simply n:m or $n \times m$ decoder ($m \leq 2^n$).
- May be used to generate 2^n (or fewer) minterms of n input variables.

Decoders

- Example: if codes 00, 01, 10, 11 are used to identify four light bulbs, we may use a 2-bit decoder:



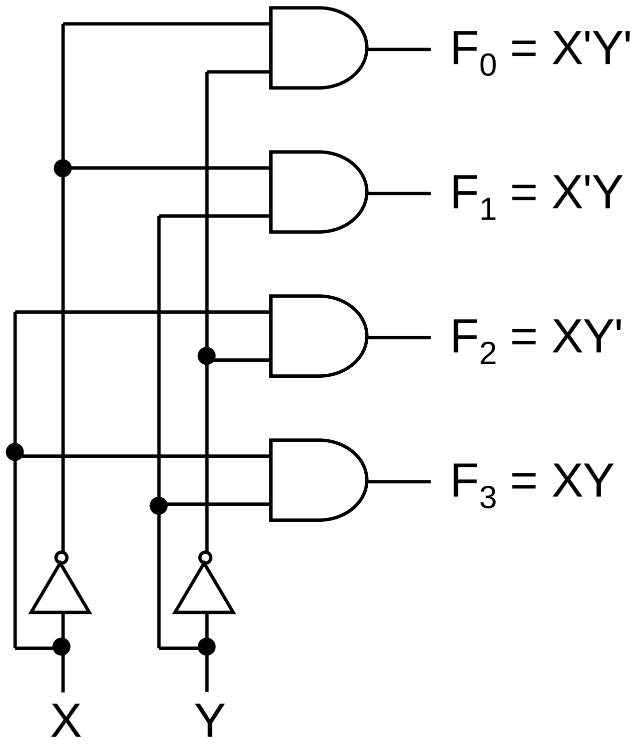
- This is a 2×4 decoder which selects an output line based on the 2-bit code supplied.
- Truth table:

X	Y	F_0	F_1	F_2	F_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Decoders

- From truth table, circuit for 2×4 decoder is:
- Note: Each output is a 2-variable minterm (**X'Y', X'Y, XY'** or **XY**)

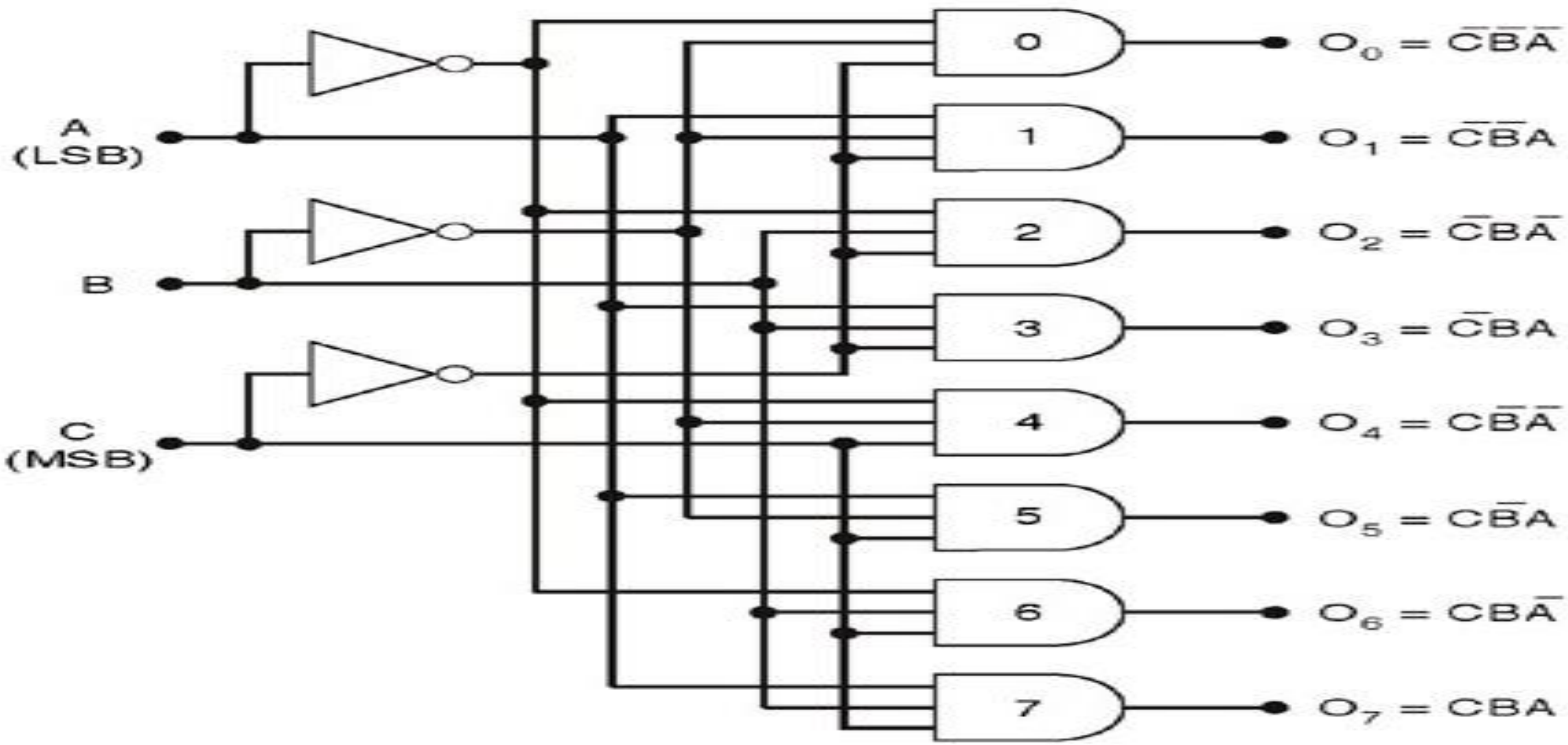
X	Y	F_0	F_1	F_2	F_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1



Decoders

- Design a 3×8 decoder by yourself.

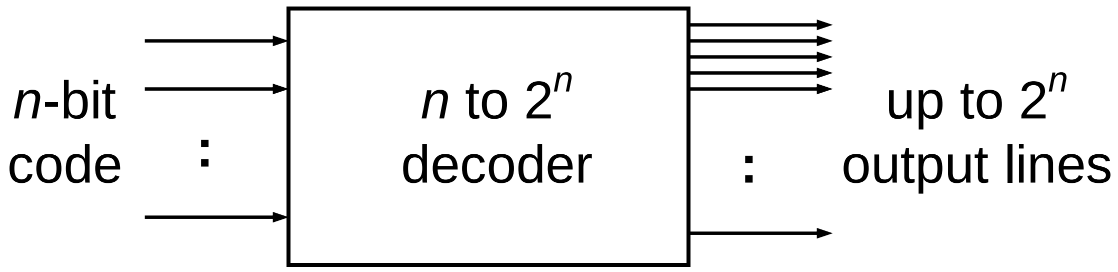
Solution



C	B	A	O_7	O_6	O_5	O_4	O_3	O_2	O_1	O_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

Decoders

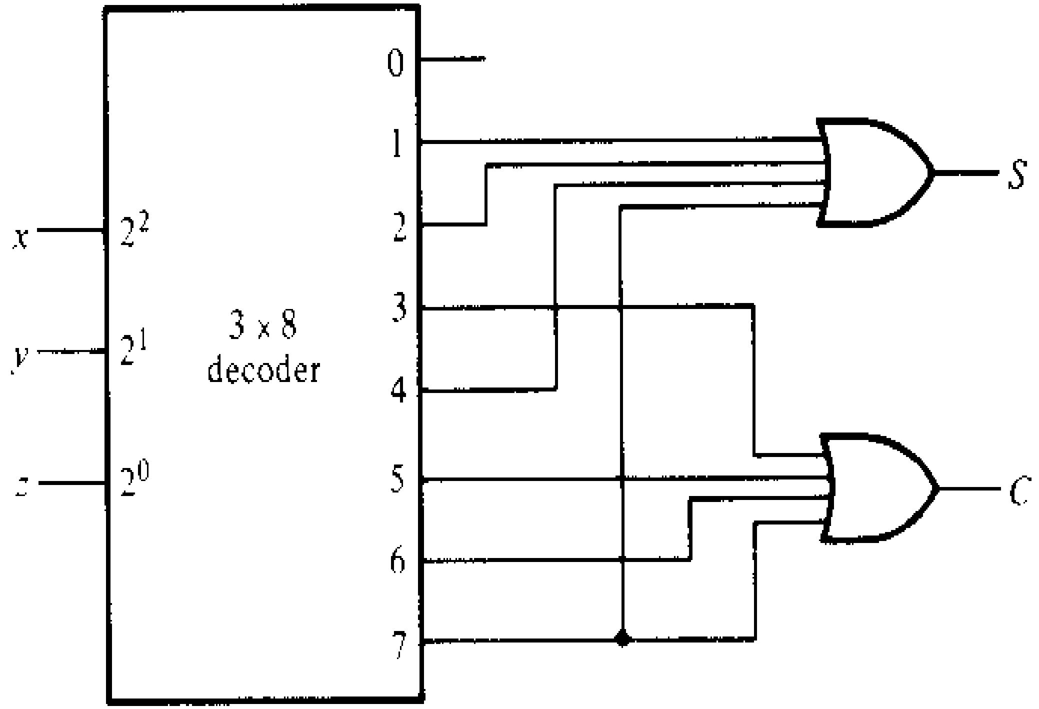
- In general, for an n-bit code, a decoder could select up to 2^n lines:



As n input generates 2^n output, which reminds us of canonical SOP, thus a decoder can be used to generate any function

Application of Decoder

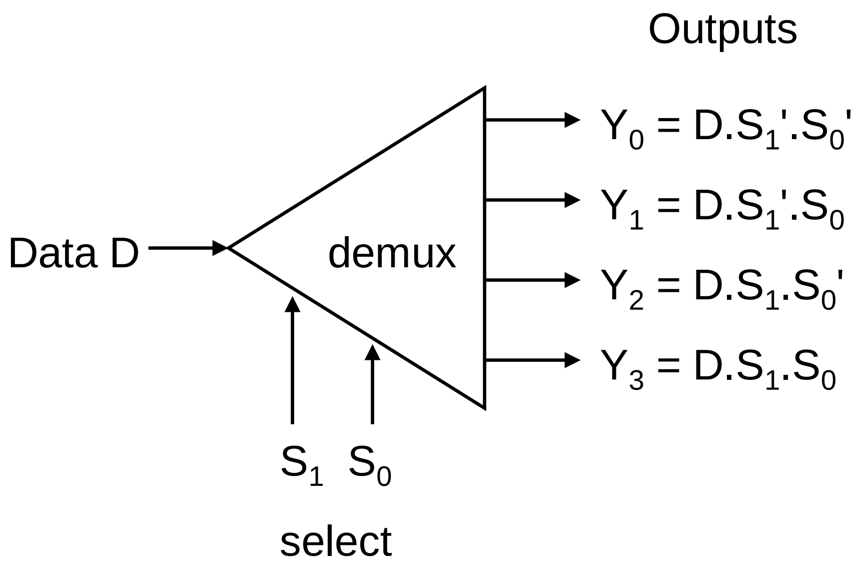
- Example 1: Full adder circuit with decoder (3 x 8 decoder)



$$S(x, y, z) = \Sigma(1, 2, 4, 7)$$
$$C(x, y, z) = \Sigma(3, 5, 6, 7)$$

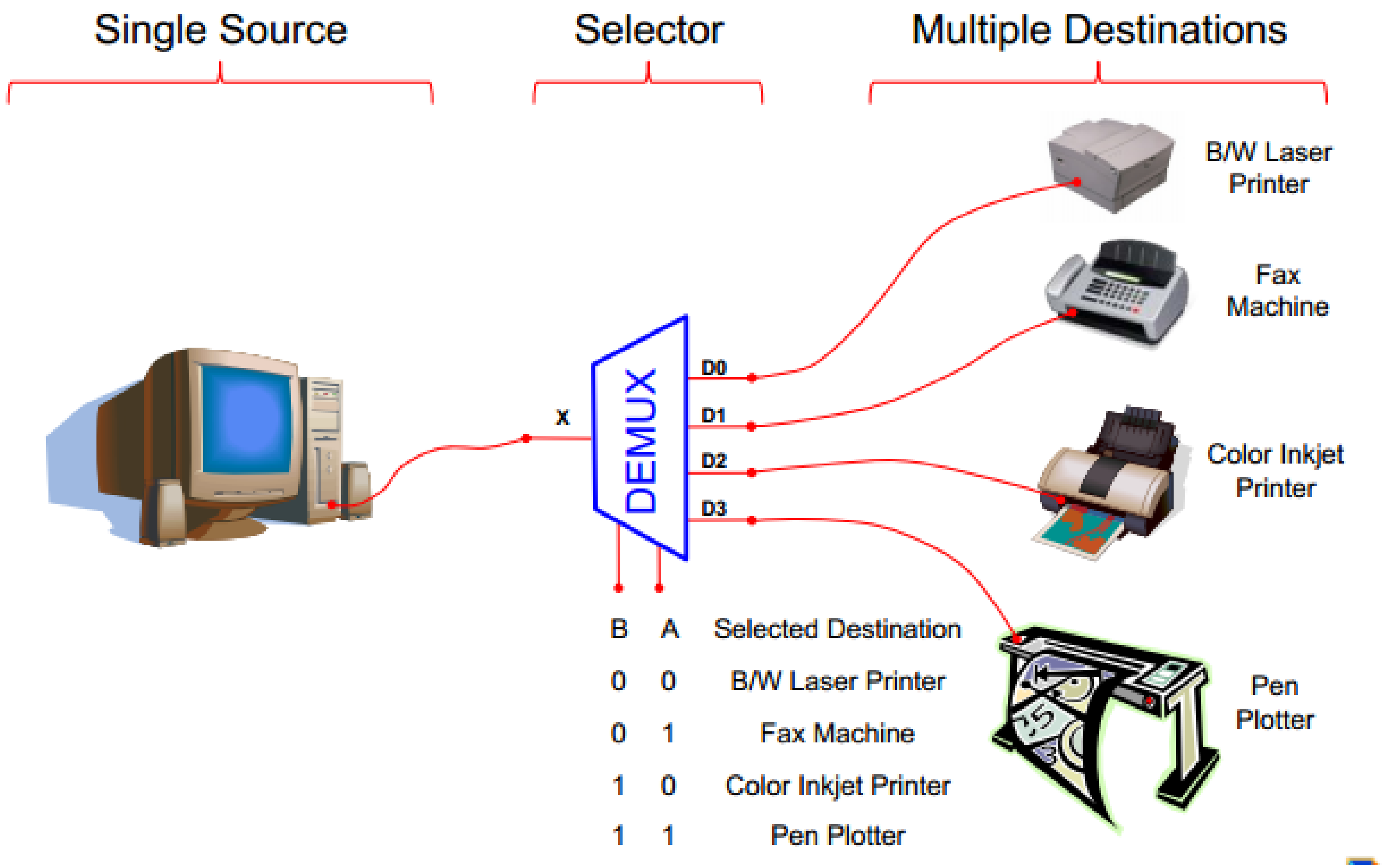
Demultiplexer

- Given an input line and a set of selection lines, the demultiplexer will direct data from input to a selected output line.
- An example of a 1-to-4 demultiplexer:



S_1	S_0	Y_0	Y_1	Y_2	Y_3
0	0	D	0	0	0
0	1	0	D	0	0
1	0	0	0	D	0
1	1	0	0	0	D

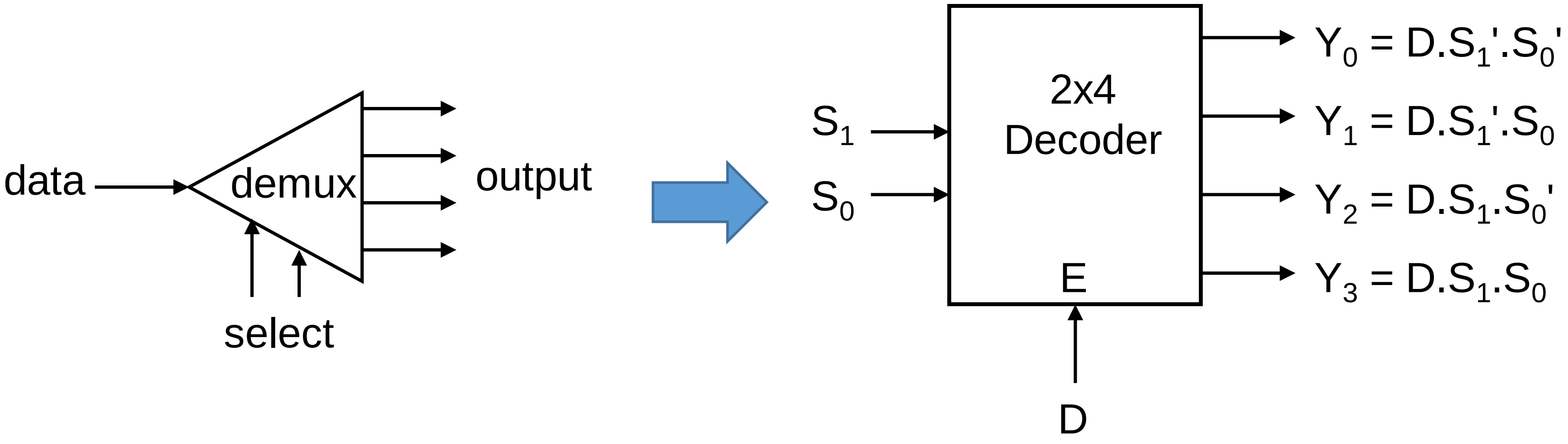
Typical Application of a DEMUX



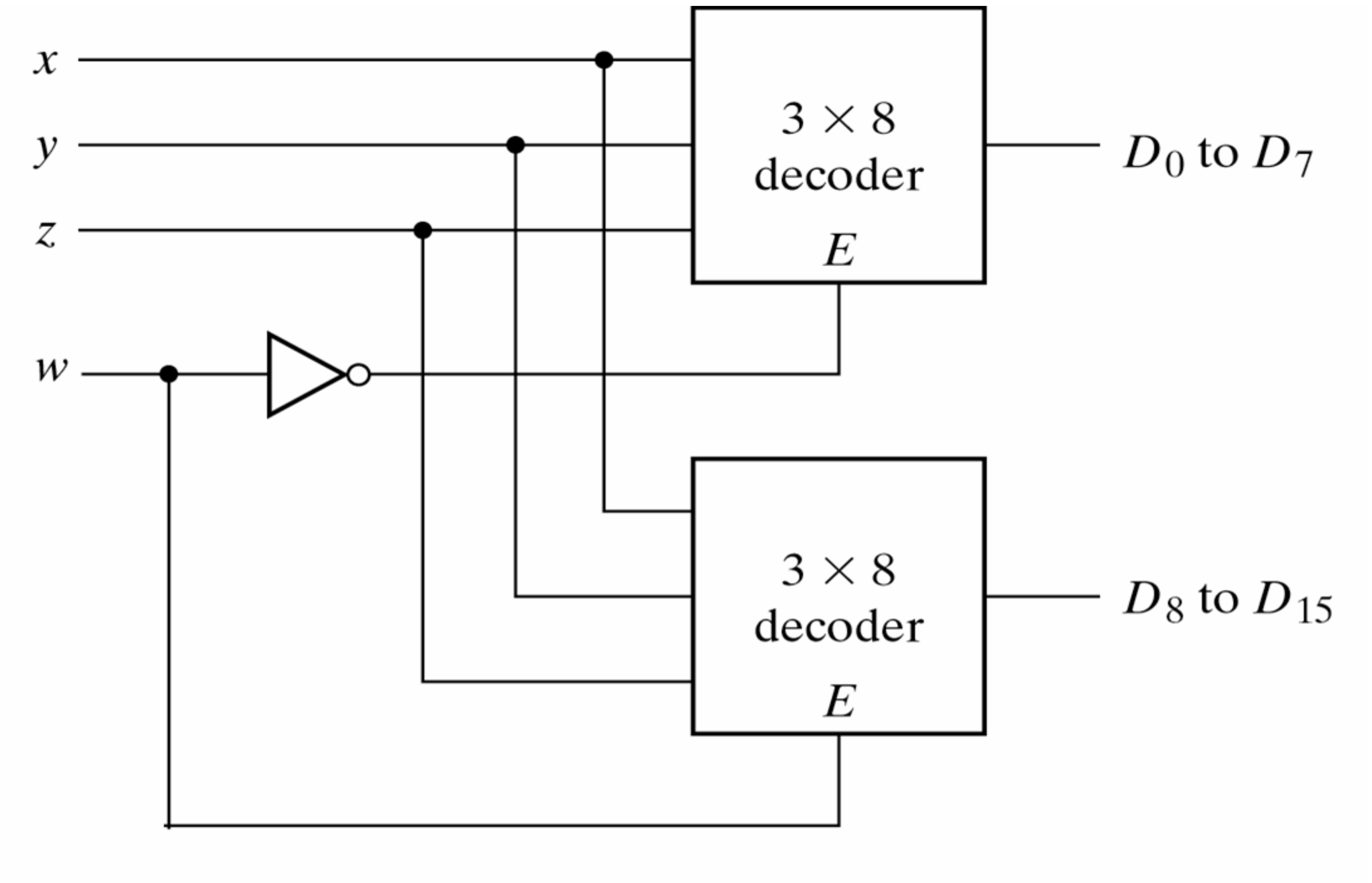
B	A	Selected Destination
0	0	B/W Laser Printer
0	1	Fax Machine
1	0	Color Inkjet Printer
1	1	Pen Plotter

Demultiplexer

- The demultiplexer is actually identical to a decoder with enable. A decoder with an enable input can function as a demultiplexer (or demux)
  Decoder + enable= demultiplexer
- The selection of a specific output line is controlled by the bit values of 'n'-selection lines.



4-line-to-16 line Decoder constructed with two 3-line-to-8 line decoders with enables



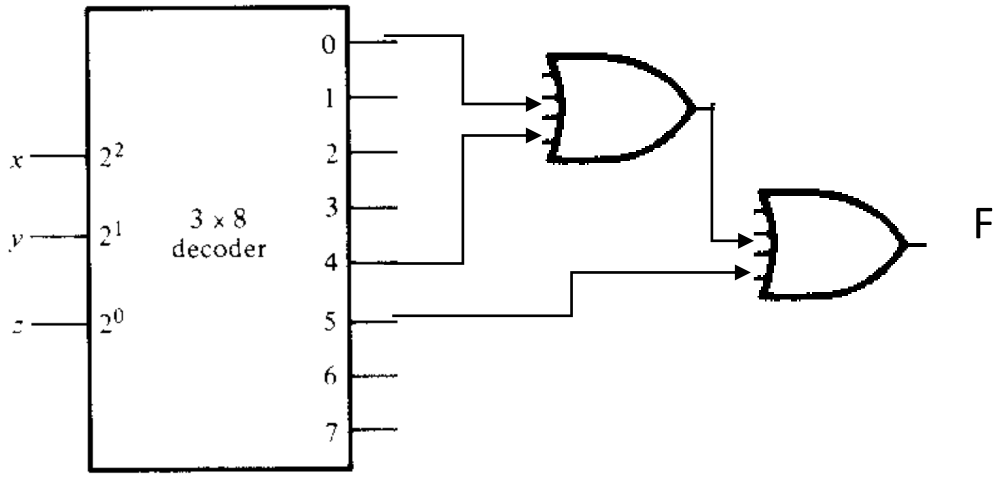
4-line-to-16 line Decoder constructed with two 3-line-to-8 line decoders with enables

W	X	Y	Z
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

- When w=0, the top decoder is enabled and the other is disabled. The bottom decoder outputs are all 0's , and the top eight outputs generate min-terms 0000 to 0111.
- When w=1, the enable conditions are reversed. The bottom decoder outputs generate min-terms 1000 to 1111, while the outputs of the top decoder are all 0's.

Implement the function

- $F=\sum(0,4,5)$

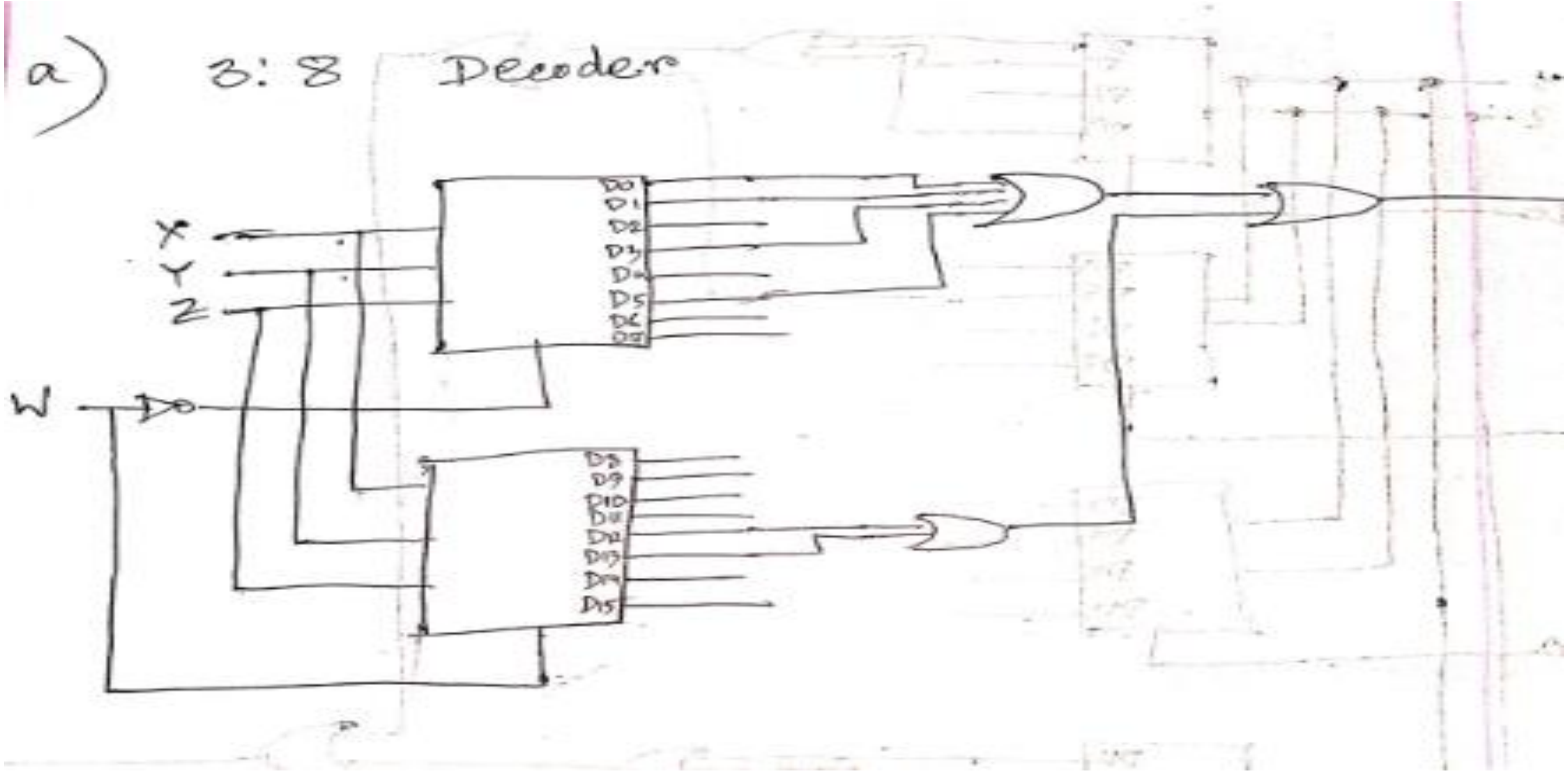


F(a,b,c,d)=∑(0,1,3,5,12,13)

Implement the above boolean function using

a. 3:8 Decoder(s).

b. 2:4 Decoder(s).



$F(a,b,c,d)=\sum(0,1,3,5,12,13)$

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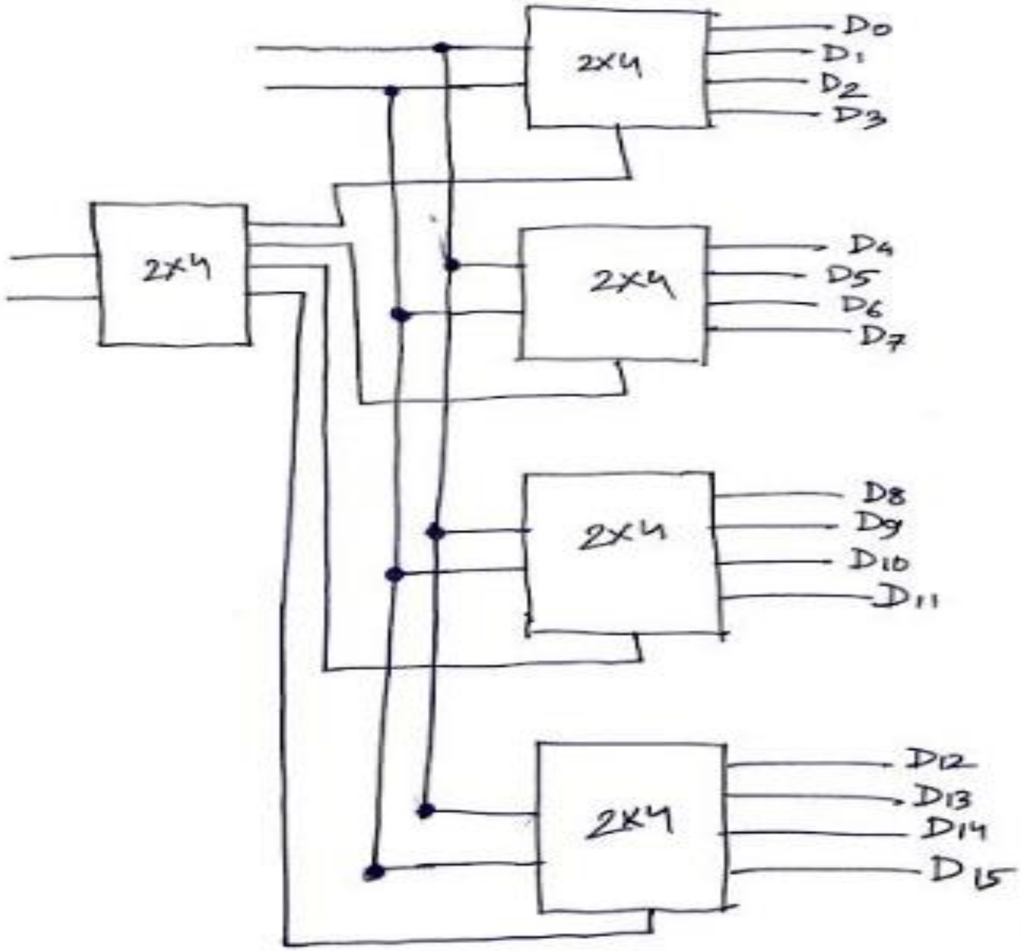

You have to connect D0,D1,D3,D5,D12 & D13 with OR Gate

Note:

- Both mux and decoder can be used to design combinational circuit.
- Decoder are mostly used to decoding binary information and mux are mostly used to select path between multiple sources and a single destination.